

**Lecture Summary for
Computer Vision A
BY
THU TA ZAW**

The fundamentals of digital image processing were introduced, including how images are represented in different color spaces such as RGB, HSB, and Lab, and how these individual channels can be leveraged for robust features extraction. Various image-processing methods including brightness and contrast normalization, histogram-based thresholding, and binarization through Otsu's method were also proposed.

Filtering techniques ranging from box and Gaussian filters to bilateral filtering, the latter of which preserves edges by combining spatial and intensity information were also discussed. Noise removal methods such as median filtering were demonstrated to effectively eliminate salt-and-pepper noise without excessively blurring important image structures.

Edge detection was explained using first-order derivative operators (such as the Sobel filter for gradient computation) and second-order derivatives (including the Laplacian and Laplacian-of-Gaussian (LoG)), showing how gradient magnitudes and zero-crossing highlight object boundaries in images.

Local feature detecting, focusing on how key points such as corners and interest points are extracted to characterize distinctive structures in images was introduced. Classical detectors such as the Moravec detector - based on intensity variation measured via Sum of Squared Differences (SSD), and the Harris Corner Detector, based on the second-moment matrix (structure tensor) and its eigen values were also covered in the lecture. The Hessian detector was also presented as a curvature-based alternative, using the determinant of the Hessian matrix to identify points of high local curvature.

FAST significantly accelerates corner detection using an optimized segment test and machine-learned decision tree. Feature descriptors such as SIFT, which provide strong scale and rotation invariance via Difference-of-Gaussian (DoG) key point detection and 128-dimensional orientation histograms, and SURF, which achieves similar invariance with higher computational efficiency through integral images and Haar-wavelet responses, were also described in the lecture.

Feature matching forms the foundation for applications such as image stitching, object recognition, and 3D reconstruction. Template matching approaches such as SSD, SAD, NCC and zero-mean NCC were also introduced along with their disadvantages.

Frequency-domain correlation methods using the 2D Fast Fourier Transform (FFT) and Phase-Only Correlation (POC) were presented, offering high precision alignment by comparing only the phase of the Fourier spectra. A significant extension involved converting images to log-polar coordinates, transforming rotation and scale variations into simple translations. This enables POC to accurately estimate rotation, scaling, and translation, making it effective for applications requiring precise registration under geometric transformations.

Panoramic image stitching was also discussed, in which corresponding feature points across images are used to compute the homography that relates one viewpoint to another. Camera intrinsic parameters (focal length, principal point) and extrinsic parameters (rotation and

translation) were explained as the basis for mapping 3D world coordinates into 2D image coordinates.

Building on camera geometry, stereo vision was introduced as a method for estimating depth by comparing images taken from slightly different viewpoints. Epipolar geometry was explained through epipolar planes, epipolar lines, and the fundamental constraint that corresponding points must lie on their respective epipolar lines. Stereo matching algorithms such as Block Matching (BM) and Semi-Global Block Matching (SGBM) were also discussed in the lecture. The real-world applications include small object detection, road-surface recognition, and high-resolution 3D reconstruction for autonomous vehicles were also mentioned in the lecture.

The machine-learning section of the lecture starts with supervised learning, introducing the perceptron as the simplest linear classifier capable of solving linearly separable tasks such as logical AND/OR operations. Multi-Layer perceptrons (MLPs) extend this to non-linear classification using activation functions such as sigmoid, tanh, and ReLU.

Backpropagation and gradient descent were discussed as the core learning mechanisms, with detailed derivations of gradient calculations using computational graphs. Several loss functions including mean squared error, mean absolute error, categorical cross-entropy, and binary cross-entropy were derived step-by-step, emphasizing how gradients are propagated to update network parameters.

Classical machine-learning models such as Support Vector Machines (SVMs), which optimize margin-based decision boundaries, and ensemble-learning methods such as AdaBoost and Random Forests, which combine multiple weak classifiers to produce robust predictions, were also introduced.

The Deep Learning Convolutional Neural Networks (CNNs) are designed to learn spatially localized features from images. Core components such as convolution layers, kernels, feature maps, pooling operations, and fully connected layers were discussed through foundational architectures including LeNet, AlexNet, GoogLeNet, and ResNet. Visualization techniques demonstrated that early layers learn low-level features like edges, while deeper layers capture increasingly complex patterns and semantic representations.

A complete MNIST classification workflow was used to demonstrate practical training steps, including input formatting, Softmax probability computation, cross-entropy loss, and backpropagation through convolutional layers. CNNs were contextualized within modern applications such as image classification, object detection, regression, and segmentation, showing how deep networks have become the standard for computer-vision tasks.

The advanced 3D reconstruction approaches such as Structure-from-Motion (SfM) and Multi-View Stereo (MVS) were also mentioned in the lecture. SfM was introduced as a pipeline for estimating camera poses and generating sparse 3D point clouds from feature correspondences across multiple images and MVS extends this with dense depth-map estimation to produce detailed 3D models, forming the basis of classical multi-view reconstruction system.

Neural Radiance Field (NeRF), which is a modern neural rendering technique that represents scenes as continuous volumetric functions, were also discussed. Using camera poses

from SfM, NeRF learns a mapping from 3D positions and view directions to color and density values. Through volume rendering and ray marching, NeRF enables photorealistic novel-view synthesis. Current active and rapidly evolving directions such as fast NeRF, dynamic scene NeRF, and low computation NeRF were also covered in the lecture.

THU TA ZAW

2415121095